

Continuous reconstruction on Banach unit balls

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Abstract

One fixed inclusion from $\ell_1([0, 1])$ to $\ell_2([0, 1])$ disproves every universal continuity-cost constant over arbitrary real Banach spaces. The construction uses nonseparable spaces and does not settle a separable variant.

Provenance and review. Separated and polished from the supplied AI-assisted research note prepared with ChatGPT. Codex reviewed the mathematical argument and accompanying checks. This is a research claim awaiting independent mathematical review; no independent human audit or formal proof verification is claimed.

1 Entry 570: no universal cost of continuous decoding

The quantification in entry 570 is over arbitrary real Banach spaces, with no separability assumption [1]. This distinction is decisive. The compact-input factor-two theorem in the cited source [2] does not apply to the following unit ball.

Write B_X for the closed unit ball. The quantity $e_n(S)$ is the infimum of $\sup_{x \in B_X} \|Sx - \varphi(N(x))\|_Y$ over continuous encoders $N : B_X \rightarrow \mathbb{R}^n$ and arbitrary decoders $\varphi : \mathbb{R}^n \rightarrow Y$. The quantity $\delta_n(S)$ is the same infimum with φ also continuous. The decoder is defined on all of $\mathbb{R}^n$, as required by the target.

Theorem 1.1. *Let*

$$I = [0, 1], \quad X = \ell_1(I), \quad Y = \ell_2(I),$$

and let $S : X \rightarrow Y$ be the canonical inclusion. Then, for every finite $n \geq 1$,

$$\delta_n(S) = 1,$$

whereas for every integer $K \geq 1$,

$$e_{2K}(S) \leq \frac{1}{\sqrt{K+1}}.$$

Therefore no universal finite constant C can satisfy $\delta_n(S) \leq Ce_n(S)$ in the stated class. In particular, $C = 2$ already fails for this operator at $n = 8$.

1.1 A continuous encoder with a discontinuous accurate decoder

Every element of $\ell_1(I)$ or $\ell_2(I)$ has countable support. The inclusion S is bounded with norm one. Fix $K \geq 1$ and put $\tau = 1/(K+1)$. Define coordinatewise soft thresholding by

$$(T_\tau x)_t = \operatorname{sgn}(x_t)(|x_t| - \tau)_+.$$

This map is 1-Lipschitz on $\ell_1(I)$. If $\|x\|_1 \leq 1$, then $T_\tau x$ has at most K nonzero coordinates: $K + 1$ coordinates strictly larger than τ in magnitude would have total ℓ_1 norm greater than one. Also,

$$\|x - T_\tau x\|_2^2 = \sum_{t \in I} \min(|x_t|, \tau)^2 \leq \tau \sum_{t \in I} |x_t| \leq \tau. \quad (1)$$

Use $2K$ real moments of the thresholded vector as the encoder:

$$N_K(x)_j = \sum_{t \in [0,1]} t^j (T_\tau x)_t, \quad j = 0, 1, \dots, 2K - 1. \quad (2)$$

Here $t^0 = 1$, including at $t = 0$. Every coordinate is continuous, indeed 1-Lipschitz in the ℓ_1 norm, because $|t^j| \leq 1$. Thus $N_K : B_X \rightarrow \mathbb{R}^{2K}$ is an admissible nonadaptive continuous encoder.

The moment map in (2) is injective on vectors supported on at most K indices. To prove this, subtract two such vectors with the same moments. Their difference has support $\{t_1, \dots, t_r\}$, where $r \leq 2K$, and its coefficients c_i satisfy

$$\sum_{i=1}^r c_i t_i^j = 0, \quad j = 0, \dots, r - 1.$$

The Vandermonde matrix of distinct nodes t_i is invertible; hence all $c_i = 0$.

Define a decoder on all of $\mathbb{R}^{2K}$ by returning the unique at-most- K -sparse vector having the given moments whenever one exists, and returning zero otherwise. No continuity is required for e_{2K} . On the encoder image it returns $T_\tau x$ exactly. Equation (1) proves

$$e_{2K}(S) \leq \sqrt{\tau} = (K + 1)^{-1/2}.$$

1.2 Every continuous decoder misses a coordinate

Let $\phi : \mathbb{R}^n \rightarrow \ell_2(I)$ be any continuous decoder. Its image is separable, since $\mathbb{R}^n$ is separable. More explicitly, choose a countable dense subset of its image and let $J \subset I$ be the union of the supports of those countably many vectors. Each support is countable, so J is countable. Since $\ell_2(J)$ is closed in $\ell_2(I)$,

$$\phi(\mathbb{R}^n) \subset \ell_2(J).$$

Choose $t \in I \setminus J$ and take the unit vector $e_t \in B_X$. For every encoder N , regardless of its form,

$$\|Se_t - \phi(N(e_t))\|_2^2 = 1 + \|\phi(N(e_t))\|_2^2 \geq 1.$$

Thus $\delta_n(S) \geq 1$. The zero decoder gives the reverse inequality, because $\|S\| = 1$.

For any putative universal C , choose $K + 1 > C^2$. Then

$$Ce_{2K}(S) \leq \frac{C}{\sqrt{K + 1}} < 1 = \delta_{2K}(S).$$

For $C = 2$, take $K = 4$, giving $2e_8(S) \leq 2/\sqrt{5} < 1$.

Remark 1.2. The same fixed operator supplies the counterexample for every proposed constant. Both its domain and codomain are nonseparable. No conclusion about the still-interesting variant restricted to separable Banach spaces is claimed.

References

- [1] M. Colbrook, AIM catalogue, [entry 570: The cost of continuous reconstruction on Banach unit balls](#). Repository statement checked 4 October 2026.
- [2] D. Krieg, E. Novak, and M. Ullrich, *How many continuous measurements are needed to learn a vector?*, [arXiv:2412.06468v2](#), 22 September 2026. Theorem 1, Section 3, Lemma 9, and the discussion of Banach unit balls.